Contraction Theory Based Control for Synchronous Machines ⋆

Myeongseok Ryu * **Kyunghwan Choi** ** **Sesun You** ***

* *Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34051, South Korea (e-mail: dding_98@kaist.ac.kr)*
** *Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34051, South Korea (e-mail: kh.choi@kaist.ac.kr)*
*** *Kyemyung University, Daegu 42601, South Korea (e-mail: yousesun@kmu.ac.kr)*

Abstract: Besides the traditional Lyapunov stability theory, contraction theory offers a interesting perspective on the stability of nonlinear systems. It allows us to analyze the stability of systems by examining the convergence of trajectories in the state space. In this paper, the contraction theory is applied to the control of synchronous machines, specifically permanent magnet synchronous motors (PMSMs). The proposed control strategy is mainly designed to ensure the contraction of the system trajectories to a desired system trajectories, under unknown (and stochastic) disturbances. Real time validation of the proposed control strategy is performed on a PMSM system. The results demonstrate the effectiveness of the proposed control strategy in achieving stability and robustness in the presence of disturbances.

Keywords: Contraction theory, permanent magnet synchronous motor (PMSM)

NOTATION

In this paper, we use the following notation:

- $:=$ denotes *defined as*.
- $(\cdot^\top)$ denotes the transpose of a matrix or a vector.
- $\boldsymbol{x} := [x_i]_{i\in\{1,\cdots,n\}} \in \mathbb{R}^n$ denotes the state vector.
- $\boldsymbol{A} := [a_{ij}]_{i,j\in\{1,\cdots,n\}} \in \mathbb{R}^{n\times n}$ denotes a matrix.
- $\lambda_i(\boldsymbol{A}),\ i \in \{\max, \min\}$ denotes the maximum and minimum singular value of $\boldsymbol{A}$, respectively.
- $\boldsymbol{I}_n$ denotes the identity matrix of size n and $\mathbf{0}_{n\times m}$ denotes the zero matrix of size $n \times m$.
- sym denotes the symmetric part of a matrix, i.e., $\mathrm{sym}(\boldsymbol{A}) := \frac{1}{2}(\boldsymbol{A} + \boldsymbol{A}^\top)$ (see Tsukamoto et al. (2021b)).

1. INTRODUCTION

1.1 Motivation

- About SM control and challenges induced by disturbances
- Advantages of contraction theory
- Recent developments in contraction theory
- Application of contraction theory to SM control

To the best of our knowledge, there has been no previous work that applies the contraction theory to the control of synchronous machines in real time. Hence, the main objective of this paper is to apply contraction theory to the control of synchronous machines and validates its effectiveness and feasibility through real-time experiments.

⋆ Sponsor and financial support acknowledgment goes here. Paper titles should be written in uppercase and lowercase letters, not all uppercase.

1.2 Contributions

The main contributions of this paper are

- A contraction theory based control strategy for synchronous machines, is proposed;
- The contraction metric is obtained by using linear matrix inequalities (LMI) which simplifies the calculation of the contraction metric;
- ...; and
- The effectiveness and the feasibility of the contraction theory based control on synchronous machines is validated through real-time experiments.

2. PRELIMINARIES

2.1 Mathematical Modeling of Synchronous Machines

Let us consider a synchronous machine presented in Lee et al. (2018), as follows:

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t}\theta = & \omega \\ \frac{\mathrm{d}}{\mathrm{d}t}\omega = & \frac{1}{J}\big(\tau - f_v\omega + \tau_L\big), \end{cases} \tag{1}$$

where $\theta \in \mathbb{R}$ is the rotor angle, $\omega \in \mathbb{R}$ is the rotor speed, $J \in \mathbb{R}_{>0}$ is the moment of inertia, $\tau := \tau(i_\alpha, i_\beta) \in \mathbb{R}^2$ is the electromagnetic torque generated by the machine, $f_v \in \mathbb{R}_{>0}$ is the viscous friction coefficient, and $\tau_L \in \mathbb{R}$ is the load torque. The electromagnetic torque $\tau \in \mathbb{R}$ is typically a function of the phase currents $i_\alpha \in \mathbb{R}$ and $i_\beta \in \mathbb{R}$ of the machine, which can be expressed as:

$$\tau = -\tfrac{3}{2}P\Phi(\sin(P\theta)i_\alpha + \cos(P\theta)i_\beta), \tag{2}$$

where $P \in \mathbb{N}$ is the number of pole pairs and $\Phi \in \mathbb{R}$ is the flux linkage.

For brevity, we will denote the system (1) as general state space form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}_1\boldsymbol{u} + \boldsymbol{B}_2 d, \tag{3}$$

where $\boldsymbol{x} := (\theta, \omega)^\top \in \mathbb{R}^2$, $\boldsymbol{u} := (i_\alpha, i_\beta)^\top \in \mathbb{R}^2$ and $d := \tau_L$ denote the state vector, the control input vector and the disturbance, respectively. The system matrix $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ and the input matrices $\boldsymbol{B}_1 := \boldsymbol{B}_1(\boldsymbol{x}) : \mathbb{R}^2 \to \mathbb{R}^{2\times 2}$ and $\boldsymbol{B}_2 \in \mathbb{R}^2$ are defined as follows:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{f_v}{J} \end{bmatrix}, \quad \boldsymbol{B}_1 = \begin{bmatrix} 0 & 0 \\ -\frac{3P\Phi}{2J}\sin(P\theta) & \frac{3P\Phi}{2J}\cos(P\theta) \end{bmatrix}, \tag{4}$$

and $\boldsymbol{B}_2 = (0, 1)^\top$. It is well known that the system (3) is controllable and observable under the assumption that the system parameters are known [MS: ..? REF].

2.2 Contraction Theory

Contraction Metrics Consider that we have the following deterministic nonlinear system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t). \tag{5}$$

The contraction theory starts with the definition of *differential dynamics* of the system (5) using *differential displacement* $\delta\boldsymbol{x}$ (Kirk, 2004, Chap 4), which is defined as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\delta\boldsymbol{x}. \tag{6}$$

The differential displacement $\delta\boldsymbol{x}$ is the infinitesimal displacement at *fixed time*.

The contraction property of the system (5) can be analyzed using the Lyapunov function $V := V(\boldsymbol{x}, t) = \frac{1}{2}\delta\boldsymbol{x}^\top \boldsymbol{M}\delta\boldsymbol{x}$, where $\boldsymbol{M} = \boldsymbol{M}^\top := \boldsymbol{M}(\boldsymbol{x}, t) \succ 0$ denotes the *contraction metric*. For the system (5) to be contracting, the following condition must hold:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}(\boldsymbol{M}\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}) \leq -2\alpha\boldsymbol{M}, \tag{7}$$

which leads to $\frac{\mathrm{d}}{\mathrm{d}t}V \leq -\alpha V$ for some $\alpha \in \mathbb{R}_{>0}$. In other words, all solutions of the system (5) are contracting to a unique solution with an exponential rate α.

On the other hand, the partial contraction theory investigates the contraction of the system using a virtual system. Following theorem details the partial contraction theory.

Theorem 1. (see Wang and Slotine (2004)). Consider a nonlinear system of the form $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{x}, t)$ and the virtual system $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q} = \boldsymbol{f}(\boldsymbol{q}, \boldsymbol{x}, t)$ are contracting with respect to $\boldsymbol{q}$. If a particular solution of virtual system verifies a smooth specific property, then all solutions of the original system verifies this property exponentially.

2.3 Perturbed Contraction System

The following theorem presents the contraction property of the perturbed system by adding a bounded perturbation $\boldsymbol{d}(\boldsymbol{x}, t)$ to the system (5):

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t) + \boldsymbol{d}(\boldsymbol{x}, t). \tag{8}$$

Theorem 2. (Eq. 15 Lohmiller and Slotine (1998)). If the system (5) is contracting with the contraction metric $\boldsymbol{M}$ which satisfies the condition (7), the path integral of the virtual system ... [MS: ADD!!]

$$\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| \leq \frac{V_l(0)}{\sqrt{\underline{m}}}\exp(-\alpha t) + \frac{\bar{d}}{\alpha}\sqrt{\frac{\overline{m}}{\underline{m}}}(1 - \exp(-\alpha t)), \tag{9}$$

[MS: ADD!!]

Proof. See Theorem 2.4 in Tsukamoto et al. (2021c).

[MS: Any more comments, if needed]

3. MAIN RESULTS

In this section, the control problem will be presented, followed by the design of the proposed control strategy based on contraction theory.

3.1 Problem Formulation

Motivated by the use of the partial contraction theory in Tsukamoto et al. (2021a), [MS: and more...], we assume that the desired trajectory (including desired control input) of the system is given without disturbances, as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}^* = \boldsymbol{A}\boldsymbol{x}^* + \boldsymbol{B}_1\boldsymbol{u}^*, \tag{10}$$

where $\boldsymbol{x}^*$ is the desired state vector and $\boldsymbol{u}^*$ is the desired control input vector.

Finally, the control objective is to design a control input $\boldsymbol{u}$ (current references) such that the system trajectories of the system (10) and the desired system (10) converges. We assume that a lower lever control is designed to track the current references $\boldsymbol{u}$, and thus we can focus on the control design for the system (3).

3.2 Control Design

Following the control design in Tsukamoto et al. (2021a), state-dependent coefficient (SDC) formulation Dani et al. (2015) is used to represent the error dynamics [MS: more details]. The SDC $\mathbb{A} := \mathbb{A}(\boldsymbol{x}, \boldsymbol{x}^*)$ is defined as follows:

$$\mathbb{A} := \int_0^1 \left[\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right]\left(c\boldsymbol{x} + (1-c)\boldsymbol{x}^*\right)\mathrm{d}c, \tag{11}$$

where $\boldsymbol{f} := \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{x}^*) = \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*) + \boldsymbol{B}_1\boldsymbol{u} - \boldsymbol{B}_1^*\boldsymbol{u}^*$ and $\boldsymbol{B}_1^* := \boldsymbol{B}_1(\boldsymbol{x}^*)$, and c is a dummy variable for the integration. The SDC for the system (5) is derived in Appendix A (see (A.2)). The existence... blah blah can be found in Dani et al. (2015), [MS: and more...].

Then, the error dynamics can be expressed as:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{x} - \boldsymbol{x}^*) =& \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*) + \boldsymbol{B}_1\boldsymbol{u} - \boldsymbol{B}_1^*\boldsymbol{u}^* + \boldsymbol{B}_2 d \\ =& \underbrace{\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*) + (\boldsymbol{B}_1 - \boldsymbol{B}_1^*)\boldsymbol{u}^*}_{=\mathbb{A}(\boldsymbol{x},\boldsymbol{x}^*)(\boldsymbol{x}-\boldsymbol{x}^*)} \\ & - \boldsymbol{B}_1(\boldsymbol{u} - \boldsymbol{u}^*) + \boldsymbol{B}_2 d \\ =& \mathbb{A}(\boldsymbol{x} - \boldsymbol{x}^*) - \boldsymbol{B}_1(\boldsymbol{u} - \boldsymbol{u}^*) + \boldsymbol{B}_2 d \\ =& \left(\mathbb{A} - \boldsymbol{B}_1\boldsymbol{R}^{-1}\boldsymbol{B}_1^\top\boldsymbol{M}\right)(\boldsymbol{x} - \boldsymbol{x}^*) + \boldsymbol{B}_2 d, \end{aligned} \tag{12}$$

with the control input $\boldsymbol{u}$ defined as:

$$\boldsymbol{u} = \boldsymbol{u}^* - \boldsymbol{R}^{-1}\boldsymbol{B}_1^\top\boldsymbol{M}(\boldsymbol{x} - \boldsymbol{x}^*) \tag{13}$$

where $\boldsymbol{R} \succ 0$ is a positive definite matrix and $\boldsymbol{M} := \boldsymbol{M}(\boldsymbol{x}, \boldsymbol{x}^*, t) \succ 0$ is the contraction metric defined in Section 2.2.1.

Finally, invoking the partial contraction theory reviewed in Section 2.2.1, we can define the virtual system as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q} = \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}^* + \left(\mathbb{A} - \boldsymbol{B}_1\boldsymbol{R}^{-1}\boldsymbol{B}_1^\top\boldsymbol{M}\right)(\boldsymbol{q} - \boldsymbol{x}^*) + \boldsymbol{B}_2 d_\mu, \tag{14}$$

where $\boldsymbol{q}$ is the virtual state vector and $d_\mu := d_\mu(d,\mu), \forall \mu \in [0,1]$ is defined as $d_\mu = d_\mu(d,0)$ and $d_\mu = d_\mu(d,1)$. The differential dynamics is then defined as:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{q} = \left(\mathbb{A} - \boldsymbol{B}_1\boldsymbol{R}^{-1}\boldsymbol{B}_1^\top\boldsymbol{M}\right)\delta\boldsymbol{q}. \tag{15}$$

Then according to Theorem 2 and the partial contraction theory, the system (14) and the system (10) will converge to the desired trajectory $\boldsymbol{x}^*$ with steady state error:

$$\boldsymbol{e}_{\mathrm{ss}} = ..?, \tag{16}$$

with the following conditions:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}(\boldsymbol{M}\mathbb{A}) + \boldsymbol{M}\boldsymbol{B}_1\boldsymbol{R}^{-1}\boldsymbol{B}_1^\top\boldsymbol{M} \leq -2\alpha\boldsymbol{M} \tag{17}$$

[MS: ADD!!]

3.3 Contraction Metric Calculation

LMI LMI

4. VALIDATIONS

4.1 Numerical Validation

4.2 Real-Time Validation

5. CONCLUSION

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Place acknowledgments here.

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Appendix A. SDC DERIVATION

In this appendix, the SDC $\mathbb{A}$ will be derived in detail. First, we follow the definition of the SDC in (11) as follows:

$$\begin{aligned}\mathbb{A} &= \int_0^1 \left[\tfrac{\partial}{\partial \boldsymbol{x}}\left(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}_1\boldsymbol{u}^*\right)\right]\left(c\boldsymbol{x} + (1-c)\boldsymbol{x}^*\right)\mathrm{d}c \\ &= \int_0^1 \left[\left(\boldsymbol{A} + \begin{bmatrix} 0 & 0 \\ b_1 & 0\end{bmatrix}\right)\right]\left(c\boldsymbol{x} + (1-c)\boldsymbol{x}^*\right)\mathrm{d}c,\end{aligned} \tag{A.1}$$

where $b_1 := b_1(\boldsymbol{x},\boldsymbol{x}^*) = -\frac{3P^2\Phi}{2J}\left(\cos(Px_1)u_1^* + \sin(Px_1)u_2^*\right)$. Then, we can rewrite the SDC as follows:

$$\mathbb{A} = \begin{bmatrix} 0 & 1 \\ b_2(\boldsymbol{x},\boldsymbol{x}^*) & -\frac{f_v}{J}\end{bmatrix}, \tag{A.2}$$

where

$$\begin{aligned} b_2(\boldsymbol{x},\boldsymbol{x}^*) :=& \tfrac{3P\Phi}{2J(x_1 - x_1^*)}\left[\{\sin(Px_1) - \sin(Px_1^*)\}\,u_1^*\right. \\ &\left. + \{\cos(Px_1) - \cos(Px_1^*)\}\,u_2^*\right]. \end{aligned}$$